

# E524064\_25

The equation  $x^4 + x + y^3 - 2y + 5 = 0$  has no integer solutions

Size  $h = 35$ , length  $l = 12.32$

**Theorem 1.** *The equation*

$$x^4 + x + y^3 - 2y + 5 = 0 \tag{1}$$

*has no integer solutions.*

The proof uses the cubic Hilbert reciprocity law over the field  $K = \mathbb{Q}(\omega)$ , where  $\omega^2 + \omega + 1 = 0$ . To a hypothetical integer solution we attach the values of 44 auxiliary polynomials  $H_0, \dots, H_{43}$  with coefficients in  $\mathbb{Z}[\omega]$ , constants  $c_0, \dots, c_{43} \in \mathbb{Z}[\omega]$ , and a sum  $B$  of cubic Hilbert symbols of these values, recorded by a graph with 187 weighted edges. By the reciprocity law, the local contributions  $B_p$  of all primes  $p$  add up to 0 modulo 3. We show that  $B_p = 0$  for  $p \notin S$ , compute  $B_p$  for the primes  $p \in S$ , and obtain a contradiction. Two standard facts from class field theory are quoted without proof: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law. Apart from these, the proof uses polynomial identities, which can be checked by expanding both sides, and finite computations with residues.

## 1 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

**The field.** Let  $K = \mathbb{Q}(\omega)$ , where  $\omega^2 + \omega + 1 = 0$ . Its ring of integers is  $\mathbb{Z}[\omega]$ , and every element of  $K$  can be written uniquely as  $A + B\omega$  with  $A, B \in \mathbb{Q}$ . We write

$$\langle A, B \rangle = A + B\omega,$$

also when  $A$  and  $B$  are polynomials in  $x$  and  $y$ . Then  $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC - BD \rangle$ , and the norm is  $N(\langle A, B \rangle) = A^2 - AB + B^2$ . In particular, a value  $\langle A, B \rangle$  with  $A, B \in \mathbb{Z}$  is zero if and only if  $A = B = 0$ .

**Places.** The finite places of  $K$  correspond to the non-zero prime ideals of  $\mathbb{Z}[\omega]$ . A prime  $p \equiv 2 \pmod{3}$  is inert: there is one place above  $p$ , with uniformizer  $p$  and residue field  $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$  with  $p^2$  elements. If  $p \equiv 1 \pmod{3}$ , the polynomial  $t^2 + t + 1$  has two roots  $r$  modulo  $p$ , and for each of them there is a place  $v_r$  above  $p$ , at which  $\omega$  is the root of  $t^2 + t + 1$  in  $\mathbb{Z}_p$  congruent to  $r$  modulo  $p$ ; the completion is  $\mathbb{Q}_p$ , the uniformizer is  $p$ , the residue field is  $\mathbb{F}_p$ , and  $\omega$  reduces to  $r$ . The prime 3 is ramified:  $\pi = 1 - \omega$  generates the unique prime ideal above 3,  $\pi^2 = -3\omega$ , and  $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$ , where  $\langle A, B \rangle$  reduces to  $A + B$ . The only infinite place of  $K$  is complex. For a finite place  $v$  we write  $K_v$  for the completion and  $v(a)$  for the normalized valuation, and we call  $a$  a *unit* at  $v$  if  $v(a) = 0$ .

**Hilbert symbols.** For a place  $v$  of  $K$  and  $a, b \in K_v^\times$ , the cubic Hilbert symbol  $(a, b)_v$  is a cube root of unity, see [1, Chapter V, §3]. We write  $(a, b)_v = \omega^{\beta_v(a, b)}$  with  $\beta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}$ , and we use the following three facts [1, Chapter V, §3 and Chapter VI, §8].

(H1)  $\beta_v(a, bc) = \beta_v(a, b) + \beta_v(a, c)$ ,  $\beta_v(a, b) = -\beta_v(b, a)$ , and  $\beta_v(a, c^3) = 0$ . In particular,  $\beta_v(a, b)$  depends only on the classes of  $a$  and  $b$  in  $K_v^\times/K_v^{\times 3}$ .

(H2) Let  $v$  be a finite place not above 3, let  $q$  be the number of elements of its residue field, and let  $\varpi$  be a uniformizer at  $v$ . Write  $a = \varpi^e u$  and  $b = \varpi^f w$  with units  $u$  and  $w$ , and define  $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$  by  $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$ , where the bar denotes the reduction to the residue field. Then  $\beta_v(a, b) = e\chi(w) - f\chi(u)$ . In particular,  $\beta_v(a, b) = 0$  if  $a$  and  $b$  are both units.

(H3) (Hilbert reciprocity.) For  $a, b \in K^\times$ ,  $\beta_v(a, b) = 0$  for all but finitely many places  $v$ , and  $\sum_v \beta_v(a, b) = 0$ , where the sum is over all places of  $K$ . At the complex place,  $\beta_v(a, b) = 0$ .

The general formula in (H2) contains an additional term  $ef\chi(-1)$ , which vanishes because  $-1 = (-1)^3$  is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values  $\beta_v$  change sign, and the proof below is the same.

**Classes at the places not above 3.** Let  $p \neq 3$  be a prime and  $v$  a place above  $p$ . For  $a = p^e u \in K_v^\times$  with a unit  $u$ , put  $c_v(a) = (e \bmod 3, \chi(u)) \in \mathbb{F}_3^2$ . By (H2) and (H1),  $c_v$  is additive,  $c_v(a)$  determines the class of  $a$  modulo cubes, and  $\beta_v(a, b) = e\chi(w) - f\chi(u)$  for  $c_v(a) = (e, \chi(u))$  and  $c_v(b) = (f, \chi(w))$ . We put  $c_p(a) = c_v(a) \in \mathbb{F}_3^2$  if  $p \equiv 2 \pmod{3}$ , and  $c_p(a) = (c_{v_r}(a), c_{v_{r'}}(a)) \in \mathbb{F}_3^4$  if  $p \equiv 1 \pmod{3}$ , where  $r < r'$  are the roots of  $t^2 + t + 1$  in  $\{1, \dots, p-1\}$ ; accordingly,  $\beta_p$  is the sum of the forms  $\beta_v$  over the places  $v$  above  $p$ . For  $p \equiv 2 \pmod{3}$  and a rational integer  $u$  prime to  $p$ , we have  $\bar{u} \in \mathbb{F}_p$ , so that  $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$  and  $\chi(u) = 0$ .

### The place above 3.

**Lemma 2.** *Let  $v$  be the place above 3. Every non-zero element of  $K_v$  is equal to  $\pi^a \omega^b 2^c (1 + 3\omega)^d$  times a cube, for unique  $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$ , and every unit congruent to 1 modulo 9 is a cube. Put  $c_3(A) = (a, b, c, d) \in \mathbb{F}_3^4$ . Then  $c_3$  is additive, and  $\beta_v(A, B) = c_3(A)^\top M_3 c_3(B)$ , where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod{3}.$$

*Proof.* For  $\alpha \in \mathbb{Z}[\omega]$ , the polynomial  $g(s) = s + 3s^2 + 3s^3 - \alpha$  satisfies  $g'(s) \equiv 1 \pmod{\pi}$ , so by Hensel's lemma it has a root  $s$  in the completion of  $\mathbb{Z}[\omega]$ , and then  $(1 + 3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$ . Hence every unit congruent to 1 modulo 9 is a cube in  $K_v$ , and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in  $\mathbb{Z}[\omega]/(9)$ , and a direct enumeration shows that the 27 products  $\omega^b 2^c (1 + 3\omega)^d$  with  $b, c, d \in \{0, 1, 2\}$  are pairwise distinct modulo 9, also after multiplication by  $-1 = (-1)^3$ . Hence they represent all 27 classes of units modulo cubes. Since  $\pi^3$  is a cube times a unit, the exponent of  $\pi$  is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute  $\beta_v$  on the 16 ordered pairs of the elements  $\pi, \omega, 2$  and  $1 + 3\omega$ . Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3),  $\beta_v$  at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to  $\bar{\omega} = 2$  and  $\bar{\omega} = 4$  in  $\mathbb{F}_7$ . The classes  $(e, \chi)$  of the four elements at these places are

|               | 2      | $\bar{\omega} = 2$ | $\bar{\omega} = 4$ |
|---------------|--------|--------------------|--------------------|
| $\pi$         | (0, 2) | (0, 0)             | (0, 2)             |
| $\omega$      | (0, 1) | (0, 2)             | (0, 2)             |
| 2             | (1, 0) | (0, 2)             | (0, 1)             |
| $1 + 3\omega$ | (0, 2) | (1, 0)             | (0, 0)             |

where, at the places above 7,  $\chi(u)$  is defined by  $\bar{u}^2 = \bar{\omega}^{\chi(u)}$  in  $\mathbb{F}_7$ , and 7 is a uniformizer. Formula (H2) and the sum over these three places give  $M_3$ .  $\square$

We write  $\beta_3$  for the form of Lemma 2. Since  $3 = -\omega^2\pi^2$  and  $4 = 2^2$ , we have  $c_3(3) = (2, 2, 0, 0)$  and  $c_3(4) = (0, 0, 2, 0)$ .

**Residue classes.** The local computations use the following description of the classes of the values of a polynomial on a residue class.

**Lemma 3.** *Let  $p$  be a prime,  $k \geq 1$ , and let  $z \in \mathbb{Z}[\omega]$  be non-zero with  $z \equiv \zeta \pmod{p^k}$ , where  $\zeta = \langle A, B \rangle$  with  $0 \leq A, B < p^k$ . Then  $c_p(z) \in b + D$  for the vector  $b$  and the subspace  $D$  given as follows.*

- (i)  $p \equiv 2 \pmod{3}$ . If  $\zeta \neq 0$ , then  $b = c_p(\zeta)$  and  $D = 0$ . If  $\zeta = 0$ , then  $b = 0$ , and  $D$  is  $\mathbb{F}_3^2$ , or the span of  $(1, 0)$  if  $z \in \mathbb{Z}$ .
- (ii)  $p \equiv 1 \pmod{3}$ . For each root  $r$ , let  $\rho$  be the root of  $t^2 + t + 1$  modulo  $p^k$  congruent to  $r$ . If  $A + B\rho \not\equiv 0 \pmod{p^k}$ , the two coordinates of  $c_{v_r}(z)$  are those of  $c_{v_r}(\zeta)$ ; otherwise these two coordinates are arbitrary.
- (iii)  $p = 3$ ,  $z \notin \mathbb{Z}$ . If  $\zeta \neq 0$ , let  $a = v(\zeta) < 2k$  be its valuation at  $\pi$  and  $m = 2k - a$ . Then  $b = c_3(\zeta)$ , and  $D = 0$  if  $m \geq 4$ , while  $D = D_m$  if  $m \leq 3$ , where  $D_1$  is spanned by  $(0, 1, 0, 0)$ ,  $(0, 0, 1, 0)$  and  $(0, 0, 0, 1)$ ,  $D_2$  by  $(0, 0, 1, 0)$  and  $(0, 0, 0, 1)$ , and  $D_3$  by  $(0, 0, 1, 1)$ . If  $\zeta = 0$ , then  $b = 0$  and  $D = \mathbb{F}_3^4$ .
- (iv)  $p = 3$ ,  $z \in \mathbb{Z}$ , so that  $B = 0$ . If  $3^k \nmid A$ , let  $j = v_3(A)$ . Then  $b = c_3(A)$  and  $D = 0$  if  $k - j \geq 2$ , while  $b = j c_3(3)$  and  $D$  is spanned by  $c_3(4)$  if  $k - j = 1$ . If  $3^k \mid A$ , then  $b = 0$  and  $D$  is spanned by  $c_3(3)$  and  $c_3(4)$ .

*Proof.* (i) If  $\zeta \neq 0$ , then  $e = v(z) = v(\zeta) < k$  and  $z/p^e \equiv \zeta/p^e \pmod{p}$ , so that  $\chi$  is determined. If  $z \in \mathbb{Z}$ , its unit part is a rational integer, and  $\chi = 0$ . (ii) At  $v_r$ ,  $z$  and  $\zeta$  are the  $p$ -adic numbers  $A' + B'\omega$  and  $A + B\omega$  with  $\omega \equiv \rho \pmod{p^k}$ , and the argument of (i) applies. (iii) If  $\zeta \neq 0$ , then  $v(z) = a$ , and  $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$ . By Lemma 2, the class of a unit depends only on its residue modulo  $9 = \omega\pi^4$ ; the units congruent to 1 modulo  $\pi^m$  form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form  $D_m$ . (iv) If  $3^k \nmid A$ , then  $z = 3^j u$  with a rational unit  $u \equiv A/3^j \pmod{3^{k-j}}$ . The class of a rational unit depends only on its residue modulo 9, the units  $\pm 1$  are cubes, and the rational units congruent to 1 modulo 3 are  $4^s$  times a unit congruent to 1 modulo 9; hence the classes are as stated, and if  $3^k \mid A$  the valuation  $j$  is also unknown.  $\square$

**Lemma 4.** *Let  $p$  be a prime, and consider vertices  $i$  with vectors  $b_i$ , subspaces  $D_i$  and vectors  $\gamma_i$ , and a set of edges  $(i, j)$  with  $i < j$  and weights  $w_{ij} \in \{1, 2\}$ . Put  $\rho_{ij} = w_{ij}$  and  $\rho_{ji} = -w_{ij}$  for the edges, and  $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$ . Suppose that  $\beta_p(d, \sigma_i) = 0$  for every  $i$  and every  $d \in D_i$ , and that  $\beta_p(d, d') = 0$  for every edge  $(i, j)$  and all  $d \in D_i, d' \in D_j$ . Then*

$$\sum_{(i,j)} w_{ij} \beta_p(c_i, c_j) + \sum_i \beta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \beta_p(b_i, b_j) + \sum_i \beta_p(b_i, \gamma_i)$$

*whenever  $c_i \in b_i + D_i$  for every  $i$ . By bilinearity, it suffices to check the hypotheses for  $d$  and  $d'$  in spanning sets of  $D_i$  and  $D_j$ .*

*Proof.* Write  $c_i = b_i + d_i$  with  $d_i \in D_i$ . As  $\beta_p$  is bilinear and  $\beta_p(u, v) = -\beta_p(v, u)$ , the left side equals the right side plus  $\sum_i \beta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \beta_p(d_i, d_j)$ , and both sums vanish.  $\square$

## 2 The graph

Put  $F(x, y) = x^4 + x + y^3 - 2y + 5$ , so that (1) is the equation  $F(x, y) = 0$ . The certificate consists of 44 polynomials  $H_i$  with coefficients in  $\mathbb{Z}[\omega]$  and constants  $c_i \in \mathbb{Z}[\omega]$ , listed in Table 1, and of 187 edges  $(i, j)$  with  $i < j$  and weights  $w_{ij} \in \{1, 2\}$ , listed below. For an edge  $(i, j)$  we put  $\rho_{ij} = w_{ij}$  and  $\rho_{ji} = -w_{ij} \bmod 3$ , and  $\rho_{ij} = 0$  if  $i$  and  $j$  are not joined by an edge. The edges  $(i, j; w_{ij})$  are (0, 7; 2), (0, 13; 1), (0, 18; 1), (0, 19; 1), (0, 25; 2), (0, 27; 2), (0, 29; 2), (0, 33; 1), (1, 8; 1), (1, 12; 1), (1, 15; 1), (1, 18; 1), (1, 19; 2), (1, 23; 2), (1, 36; 1), (1, 40; 2), (1, 42; 2), (2, 8; 1), (2, 9; 1), (2, 23; 1), (2, 25; 1), (2, 26; 1), (2, 27; 2), (2, 29; 1), (2, 37; 1), (2, 38; 2), (2, 40; 1), (2, 43; 2), (3, 8; 2), (3, 10; 2), (3, 17; 2), (3, 18; 2), (3, 19; 2), (3, 21; 1), (3, 23; 1), (3, 27; 1), (3, 29; 1), (3, 31; 2), (3, 35; 1), (3, 36; 1), (3, 37; 1), (3, 40; 1), (4, 8; 1), (4, 14; 1), (4, 17; 1), (4, 25; 1), (4, 28; 2), (4, 36; 1), (4, 39; 2), (4, 43; 2), (5, 8; 1), (5, 16; 1), (5, 17; 2), (5, 18; 1), (5, 28; 2), (5, 29; 1), (5, 31; 1), (5, 35; 1), (5, 36; 1), (5, 38; 2), (5, 43; 2), (6, 9; 1), (6, 15; 2), (6, 17; 2), (6, 18; 1), (6, 25; 2), (6, 40; 1), (7, 16; 2), (7, 32; 2), (7, 37; 1), (7, 41; 1), (7, 42; 2), (8, 10; 1), (8, 12; 1), (8, 37; 2), (9, 12; 1), (9, 13; 1), (9, 18; 1), (9, 24; 2), (9, 29; 2), (9, 42; 2), (10, 12; 1), (10, 13; 2), (10, 21; 2), (10, 27; 2), (10, 30; 2), (10, 35; 2), (10, 38; 2), (10, 43; 2), (11, 15; 1), (11, 17; 2), (11, 40; 2), (11, 41; 2), (12, 13; 2), (12, 14; 2), (12, 15; 1), (12, 19; 2), (12, 21; 1), (12, 26; 2), (12, 31; 1), (12, 35; 2), (12, 37; 1), (12, 38; 1), (12, 40; 1), (13, 18; 2), (13, 20; 2), (13, 23; 2), (13, 24; 2), (13, 40; 1), (13, 42; 1), (14, 18; 2), (14, 20; 1), (14, 25; 1), (14, 26; 1), (14, 31; 2), (14, 36; 1), (14, 38; 2), (15, 19; 2), (15, 20; 2), (15, 24; 1), (15, 25; 1), (15, 32; 1), (15, 39; 1), (16, 17; 1), (16, 24; 1), (16, 30; 1), (17, 26; 2), (17, 27; 1), (18, 23; 1), (18, 24; 1), (18, 28; 1), (18, 36; 1), (18, 37; 2), (18, 40; 2), (18, 43; 2), (19, 28; 2), (19, 29; 2), (19, 32; 2), (19, 42; 1), (19, 43; 2), (20, 22; 2), (20, 32; 2), (20, 37; 1), (20, 38; 2), (20, 41; 1), (21, 35; 1), (21, 38; 1), (22, 25; 2), (22, 28; 1), (22, 32; 2), (22, 37; 1), (22, 39; 2), (22, 40; 1), (23, 24; 1), (23, 33; 2), (23, 40; 1), (24, 27; 2), (24, 29; 1), (24, 30; 1), (24, 31; 2), (24, 36; 1), (24, 39; 2), (24, 40; 1), (25, 26; 2), (25, 32; 2), (25, 34; 1), (25, 39; 1), (25, 41; 1), (26, 34; 1), (27, 30; 1), (27, 31; 1), (27, 32; 1), (27, 34; 1), (28, 39; 2), (29, 31; 2), (29, 37; 2), (29, 41; 1), (30, 32; 1), (30, 33; 2), (30, 36; 1), (32, 33; 1), (32, 41; 2), (33, 35; 1), (34, 36; 2), (34, 39; 1), (35, 38; 2), (35, 40; 1). Let  $(\xi, \eta)$  denote  $(x, y)$  or  $(y, x)$ . We call a vertex  $i$  *linear* if  $H_i = \varepsilon_i(\eta - r_i(\xi))$  for a unit  $\varepsilon_i$  of  $\mathbb{Z}[\omega]$  and a polynomial  $r_i$  with coefficients in  $\mathbb{Z}[\omega]$ ; the last column of Table 1 gives  $\eta = r_i(\xi)$  for the linear vertices. For a linear vertex  $i$ , we put  $f_i(t) = F$  and  $g_{ij}(t) = H_j$  evaluated at  $\xi = t, \eta = r_i(t)$ ; these are polynomials in  $t$  with coefficients in  $\mathbb{Z}[\omega]$ . For every linear vertex  $i$ , Table 2 gives an integer  $D_i \geq 1$  and a polynomial  $w_i(t)$  with coefficients in  $\mathbb{Z}[\omega]$  such that

$$D_i^3 c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^3 = f_i Q_i \quad (2)$$

for a polynomial  $Q_i(t)$  with coefficients in  $\mathbb{Z}[\omega]$ , where  $\rho_{ij} \in \{0, 1, 2\}$ ;  $Q_i$  is the quotient of the division of the left side by  $f_i$ , and the remainder is zero. For every edge  $(i, j)$  with a linear endpoint, Table 3 gives the first linear endpoint  $k \in \{i, j\}$  and the factorization of the norm of the resultant  $R_{ij} = \text{Res}_t(f_k, g_{kl})$ , where  $\{k, l\} = \{i, j\}$ ; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using  $\omega^2 = -1 - \omega$ .

## 3 Proof of Theorem 1

Suppose that  $(x, y) \in \mathbb{Z}^2$  is a solution of (1), so that  $F(x, y) = 0$ ; from now on the polynomials denote their values at  $(x, y)$ , which lie in  $\mathbb{Z}[\omega]$ .

**Step 1: the values are non-zero.** For  $i \in \{0, 1, 2, 3, 4, 6, 9, 12, 13, 14, 18, 19, 20, 23, 25, 26, 28, 29, 30, 31, 32, 34, 36, 37, 41, 42\}$ , there is no residue pair  $(a, b)$  modulo 2 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{2}$ , as one checks for the  $2^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{15, 17, 22, 24, 27, 39, 43\}$ , there is no residue pair  $(a, b)$  modulo 3 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0$

(mod 3), as one checks for the  $3^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{7, 16, 21, 35, 38, 40\}$ , there is no residue pair  $(a, b)$  modulo 4 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{4}$ , as one checks for the  $4^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{5, 10\}$ , there is no residue pair  $(a, b)$  modulo 5 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{5}$ , as one checks for the  $5^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{33\}$ , there is no residue pair  $(a, b)$  modulo 7 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{7}$ , as one checks for the  $7^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{8, 11\}$ , there is no residue pair  $(a, b)$  modulo 11 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{11}$ , as one checks for the  $11^2$  residue pairs; hence  $H_i \neq 0$ . Also, all  $c_i$  are non-zero.

**Step 2: reciprocity.** For every place  $v$  of  $K$ , put

$$B_v = \sum_{(i,j)} w_{ij} \beta_v(H_i, H_j) + \sum_i \beta_v(H_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \quad (3)$$

the first sum over the edges, and, for a rational prime  $p$ , put  $B_p = \sum_{v|p} B_v$ . Applying (H3) to every pair  $(H_i, H_j)$  with an edge, multiplied by  $w_{ij}$ , and to every pair  $(H_i, c_i)$ , and adding, we find that  $B_v = 0$  for all but finitely many  $v$  and  $\sum_p B_p = 0$ , as the complex place contributes 0. By (H1) and the definition of  $c_p$ ,  $B_p = \sum_{(i,j)} w_{ij} \beta_p(c_p(H_i), c_p(H_j)) + \sum_i \beta_p(c_p(H_i), c_p(c_i))$ , where  $\beta_p$  is the form of Lemma 2 if  $p = 3$ .

**Step 3: the primes outside  $S$ .** Let  $S = \{2, 3, 7, 13\}$ . It contains 3 and the primes dividing the norms of the constants  $c_i$ , of the  $D_i$ , of the  $R_{ij}$ . Let  $p \notin S$ , and let  $v$  be a place above  $p$ , with residue field  $k_v$ ; then all these numbers are units at  $v$ . First, no edge  $(i, j)$  has both values divisible by  $v$ . Indeed, suppose that  $v$  divides  $H_i$  and  $H_j$ . If  $R_{ij}$  is defined, let  $k$  be the linear endpoint used for it and  $\{k, l\} = \{i, j\}$ , and put  $t = \xi$ . As  $H_k = \varepsilon_k(\eta - r_k(\xi))$ , we have  $\eta \equiv r_k(t)$  modulo  $v$ , hence  $f_k(t) \equiv F(x, y) = 0$  and  $g_{kl}(t) \equiv H_l \equiv 0$  modulo  $v$ . The resultant can be written as  $R_{ij} = Af_k + Cg_{kl}$  with polynomials  $A$  and  $C$  with coefficients in  $\mathbb{Z}[\omega]$ , so  $v$  divides  $R_{ij}$ , a contradiction. By (H2), a term of (3) in which all arguments are units vanishes. Now let  $v$  divide  $H_i$ . Collecting the terms of (3) which contain  $H_i$ , and using (H1) and  $\rho_{ji} = -w_{ij}$  for the edges  $(j, i)$  with  $j < i$ , we obtain  $\beta_v(H_i, \Pi_i)$  with  $\Pi_i = c_i \prod_j H_j^{\rho_{ij}}$ , which is a unit. We claim that  $D_i^3 \Pi_i \equiv W^3$  modulo  $v$  for some  $W \in \mathbb{Z}[\omega]$ . If  $i$  is linear, put  $t = \xi$ ; then  $g_{ij}(t) \equiv H_j$  and  $f_i(t) \equiv 0$  modulo  $v$  as above, and (2) gives the claim with  $W = w_i(t)$ . Hence the reduction of  $\Pi_i$  is a non-zero cube in  $k_v$ ,  $\chi(\Pi_i) = 0$ , and  $\beta_v(H_i, \Pi_i) = v(H_i)\chi(\Pi_i) = 0$  by (H2). Hence  $B_v = 0$ , and  $B_p = 0$ .

**Step 4: the primes in  $S$ .** Let  $p \in S$ . Here 2 is inert, the places above 7 are  $v_2$  and  $v_4$ , and the places above 13 are  $v_3$  and  $v_9$ . Suppose that  $x \equiv a$  and  $y \equiv b \pmod{p^k}$ . Then  $H_i \equiv H_i(a, b) \pmod{p^k}$ , and Lemma 3, applied with  $\zeta$  the residue of  $H_i(a, b)$ , gives a vector  $b_i$  and a subspace  $D_i$  with  $c_p(H_i) \in b_i + D_i$ ; when  $H_i$  has rational coefficients,  $H_i \in \mathbb{Z}$ , and we use this in (i) and (iv). With  $\gamma_i = c_p(c_i)$ , if the hypotheses of Lemma 4 hold, then  $B_p$  is determined by  $a$ ,  $b$  and  $k$ . Starting from the solutions of (1) modulo  $p$ , we refine a residue class  $x \equiv a$ ,  $y \equiv b \pmod{p^k}$  to the classes modulo  $p^{k+1}$  that contain solutions of (1) modulo  $p^{k+1}$ , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes and the value of  $B_p$  in each; every integer solution lies in one of them. The result is  $B_2 = 2$ ,  $B_3 = 2$ ,  $B_7 = 0$ ,  $B_{13} = 0$ .

**Step 5: contradiction.** By Steps 2 and 3,  $\sum_{p \in S} B_p = 0$ . But by Step 4,  $\sum_{p \in S} B_p = 1$  in  $\mathbb{Z}/3\mathbb{Z}$ . This contradiction proves the theorem.  $\square$

## References

- [1] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.

Table 1: The polynomials  $H_i$ , the constants  $c_i$  and, for the linear vertices,  $\eta = r_i(\xi)$ . Here  $\langle A, B \rangle = A + B\omega$ .

| $i$ | $H_i$                                        | $c_i$                    | chart                                        |
|-----|----------------------------------------------|--------------------------|----------------------------------------------|
| 0   | $\langle -x^2 - y + 1, -x^2 + x + 1 \rangle$ | $\langle 1, 0 \rangle$   | $y = \langle -x^2 + 1, -x^2 + x + 1 \rangle$ |
| 1   | $\langle x^2 - y + 1, x^2 - 1 \rangle$       | $\langle -1, -1 \rangle$ | $y = \langle x^2 + 1, x^2 - 1 \rangle$       |
| 2   | $\langle x + 1, 1 \rangle$                   | $\langle 1, 0 \rangle$   | $x = \langle -1, -1 \rangle$                 |
| 3   | $\langle x + 1, -1 \rangle$                  | $\langle 20, 32 \rangle$ | $x = \langle -1, 1 \rangle$                  |
| 4   | $\langle x, -1 \rangle$                      | $\langle 0, -1 \rangle$  | $x = \langle 0, 1 \rangle$                   |
| 5   | $\langle x - 1, 0 \rangle$                   | $\langle 7, 0 \rangle$   | $x = \langle 1, 0 \rangle$                   |
| 6   | $\langle x - 1, -1 \rangle$                  | $\langle 2, 0 \rangle$   | $x = \langle 1, 1 \rangle$                   |
| 7   | $\langle x + y + 1, 0 \rangle$               | $\langle 4, 0 \rangle$   | $y = \langle -x - 1, 0 \rangle$              |
| 8   | $\langle x + y - 1, 0 \rangle$               | $\langle 4, 0 \rangle$   | $y = \langle -x + 1, 0 \rangle$              |
| 9   | $\langle x - y, -y \rangle$                  | $\langle 4, 0 \rangle$   | $y = \langle 0, -x \rangle$                  |
| 10  | $\langle -y + 1, 0 \rangle$                  | $\langle 1, 0 \rangle$   | $y = \langle 1, 0 \rangle$                   |
| 11  | $\langle -y - 1, 0 \rangle$                  | $\langle 2, -2 \rangle$  | $y = \langle -1, 0 \rangle$                  |
| 12  | $\langle x + 2, 1 \rangle$                   | $\langle -4, 8 \rangle$  | $x = \langle -2, -1 \rangle$                 |
| 13  | $\langle -x - y + 1, 1 \rangle$              | $\langle -1, -1 \rangle$ | $y = \langle -x + 1, 1 \rangle$              |
| 14  | $\langle x - y + 1, 1 \rangle$               | $\langle 2, -2 \rangle$  | $y = \langle x + 1, 1 \rangle$               |
| 15  | $\langle -y + 1, -x \rangle$                 | $\langle 2, 0 \rangle$   | $y = \langle 1, -x \rangle$                  |
| 16  | $\langle x + 2, -y + 1 \rangle$              | $\langle 2, 0 \rangle$   | $y = \langle -x - 1, -x - 2 \rangle$         |
| 17  | $\langle x^2 + x - y + 1, x^2 \rangle$       | $\langle 4, 0 \rangle$   | $y = \langle x^2 + x + 1, x^2 \rangle$       |
| 18  | $\langle -y, -1 \rangle$                     | $\langle -2, -2 \rangle$ | $y = \langle 0, -1 \rangle$                  |
| 19  | $\langle -y + 1, 1 \rangle$                  | $\langle -3, -3 \rangle$ | $y = \langle 1, 1 \rangle$                   |
| 20  | $\langle -y - 1, -1 \rangle$                 | $\langle -2, -2 \rangle$ | $y = \langle -1, -1 \rangle$                 |
| 21  | $\langle x - y, 0 \rangle$                   | $\langle 2, 0 \rangle$   | $y = \langle x, 0 \rangle$                   |
| 22  | $\langle x + 2, 2 \rangle$                   | $\langle 0, -1 \rangle$  | $x = \langle -2, -2 \rangle$                 |
| 23  | $\langle -x - y + 1, -x - 1 \rangle$         | $\langle -2, -2 \rangle$ | $y = \langle -x + 1, -x - 1 \rangle$         |
| 24  | $\langle x, -2 \rangle$                      | $\langle -2, 4 \rangle$  | $x = \langle 0, 2 \rangle$                   |
| 25  | $\langle -y + 2, 1 \rangle$                  | $\langle 1, 0 \rangle$   | $y = \langle 2, 1 \rangle$                   |
| 26  | $\langle x, -y \rangle$                      | $\langle 0, -2 \rangle$  | $y = \langle -x, -x \rangle$                 |
| 27  | $\langle x, y - 1 \rangle$                   | $\langle 4, 0 \rangle$   | $y = \langle x + 1, x \rangle$               |
| 28  | $\langle x, y \rangle$                       | $\langle 4, 0 \rangle$   | $y = \langle x, x \rangle$                   |
| 29  | $\langle x - y, -1 \rangle$                  | $\langle 4, -4 \rangle$  | $y = \langle x, -1 \rangle$                  |
| 30  | $\langle x - y + 2, 1 \rangle$               | $\langle 4, 0 \rangle$   | $y = \langle x + 2, 1 \rangle$               |
| 31  | $\langle x + y - 1, y \rangle$               | $\langle 4, -4 \rangle$  | $y = \langle 0, x - 1 \rangle$               |
| 32  | $\langle -x - y, x^2 - x - 1 \rangle$        | $\langle 1, 0 \rangle$   | $y = \langle -x, x^2 - x - 1 \rangle$        |
| 33  | $\langle x, 0 \rangle$                       | $\langle -4, -4 \rangle$ | $x = \langle 0, 0 \rangle$                   |
| 34  | $\langle x, 1 \rangle$                       | $\langle 2, 0 \rangle$   | $x = \langle 0, -1 \rangle$                  |
| 35  | $\langle x, y + 1 \rangle$                   | $\langle 0, -6 \rangle$  | $y = \langle x - 1, x \rangle$               |
| 36  | $\langle x + 1, y \rangle$                   | $\langle 1, -1 \rangle$  | $y = \langle x + 1, x + 1 \rangle$           |
| 37  | $\langle x + y, y \rangle$                   | $\langle 0, -2 \rangle$  | $y = \langle 0, x \rangle$                   |
| 38  | $\langle x - y - 1, -y - 1 \rangle$          | $\langle 2, 0 \rangle$   | $y = \langle -1, -x \rangle$                 |
| 39  | $\langle x - y + 1, 0 \rangle$               | $\langle 2, 0 \rangle$   | $y = \langle x + 1, 0 \rangle$               |
| 40  | $\langle -y - 1, x + 1 \rangle$              | $\langle 4, 0 \rangle$   | $y = \langle -1, x + 1 \rangle$              |
| 41  | $\langle x - 1, 1 \rangle$                   | $\langle 0, -4 \rangle$  | $x = \langle 1, -1 \rangle$                  |
| 42  | $\langle x - 2, -1 \rangle$                  | $\langle 0, -1 \rangle$  | $x = \langle 2, 1 \rangle$                   |
| 43  | $\langle -x - y, 0 \rangle$                  | $\langle 4, 0 \rangle$   | $y = \langle -x, 0 \rangle$                  |

Table 2: The data of the identities (2); the entry  $j:e$  means  $\rho_{ij} = e \neq 0$ .

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| $i = 0, D_i = 1, \rho_{ij}: 7:2, 13:1, 18:1, 19:1, 25:2, 27:2, 29:2, 33:1$                      |
| $w_i \quad \langle t^5 - t^4 - 2t^3 + t^2 + t, \quad t^5 - 3t^4 + 3t^2 - t \rangle$             |
| $i = 1, D_i = 1, \rho_{ij}: 8:1, 12:1, 15:1, 18:1, 19:2, 23:2, 36:1, 40:2, 42:2$                |
| $w_i \quad \langle -t^5 - 2t^4 + 7t^3 - 6t^2 - 5t, \quad -3t^4 + 5t^3 + 3t^2 - 10t + 4 \rangle$ |
| $i = 2, D_i = 1, \rho_{ij}: 8:1, 9:1, 23:1, 25:1, 26:1, 27:2, 29:1, 37:1, 38:2, 40:1, 43:2$     |
| $w_i \quad \langle -4t + 8, \quad \quad \quad \rangle$                                          |

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|                                                                                                                     |                                                                                         |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 8)                                                                                                                  |                                                                                         |
| $i = 3, D_i = 1, \rho_{ij}: 8:2, 10:2, 17:2, 18:2, 19:2, 21:1, 23:1, 27:1, 29:1, 31:2, 35:1, 36:1, 37:1, 40:1$      |                                                                                         |
| $w_i$                                                                                                               | $\langle -144t^2 + 224t + 400,$<br>$16t^2 - 224t + 640 \rangle$                         |
| $i = 4, D_i = 1, \rho_{ij}: 8:1, 14:1, 17:1, 25:1, 28:2, 36:1, 39:2, 43:2$                                          |                                                                                         |
| $w_i$                                                                                                               | $\langle 2t^2 - 6t + 4,$<br>$-2t + 2 \rangle$                                           |
| $i = 5, D_i = 1, \rho_{ij}: 8:1, 16:1, 17:2, 18:1, 28:2, 29:1, 31:1, 35:1, 36:1, 38:2, 43:2$                        |                                                                                         |
| $w_i$                                                                                                               | $\langle 14t^2 - 56t + 42,$<br>$-14t + 14 \rangle$                                      |
| $i = 6, D_i = 1, \rho_{ij}: 9:1, 15:2, 17:2, 18:1, 25:2, 40:1$                                                      |                                                                                         |
| $w_i$                                                                                                               | $\langle -2t + 4,$<br>$-2t^2 + 6t - 4 \rangle$                                          |
| $i = 7, D_i = 1, \rho_{ij}: 0:1, 16:2, 32:2, 37:1, 41:1, 42:2$                                                      |                                                                                         |
| $w_i$                                                                                                               | $\langle -2t^3 + 2t + 6,$<br>$-4t^2 + 4t + 6 \rangle$                                   |
| $i = 8, D_i = 1, \rho_{ij}: 1:2, 2:2, 3:1, 4:2, 5:2, 10:1, 12:1, 37:2$                                              |                                                                                         |
| $w_i$                                                                                                               | $\langle 0,$<br>$5t^3 + 3t^2 + 3t - 4 \rangle$                                          |
| $i = 9, D_i = 1, \rho_{ij}: 2:2, 6:2, 12:1, 13:1, 18:1, 24:2, 29:2, 42:2$                                           |                                                                                         |
| $w_i$                                                                                                               | $\langle 6t^3 - 2t^2 + 6t - 8,$<br>$4t^3 - 6t^2 - 2t - 14 \rangle$                      |
| $i = 10, D_i = 1, \rho_{ij}: 3:1, 8:2, 12:1, 13:2, 21:2, 27:2, 30:2, 35:2, 38:2, 43:2$                              |                                                                                         |
| $w_i$                                                                                                               | $\langle -4t^3 - 2t^2 - 2t + 8,$<br>$0 \rangle$                                         |
| $i = 11, D_i = 1, \rho_{ij}: 15:1, 17:2, 40:2, 41:2$                                                                |                                                                                         |
| $w_i$                                                                                                               | $\langle -2t + 4,$<br>$-4t - 4 \rangle$                                                 |
| $i = 12, D_i = 1, \rho_{ij}: 1:2, 8:2, 9:2, 10:2, 13:2, 14:2, 15:1, 19:2, 21:1, 26:2, 31:1, 35:2, 37:1, 38:1, 40:1$ |                                                                                         |
| $w_i$                                                                                                               | $\langle -80t^2 + 444t + 1148,$<br>$-36t^2 + 932t + 56 \rangle$                         |
| $i = 13, D_i = 1, \rho_{ij}: 0:2, 9:2, 10:1, 12:1, 18:2, 20:2, 23:2, 24:2, 40:1, 42:1$                              |                                                                                         |
| $w_i$                                                                                                               | $\langle -65t^2 - 55t - 10,$<br>$-29t^3 - 50t^2 + 25t + 34 \rangle$                     |
| $i = 14, D_i = 1, \rho_{ij}: 4:2, 12:1, 18:2, 20:1, 25:1, 26:1, 31:2, 36:1, 38:2$                                   |                                                                                         |
| $w_i$                                                                                                               | $\langle -4t^3 + 8t^2 + 10t + 6,$<br>$-2t^3 - 2t^2 + 8t \rangle$                        |
| $i = 15, D_i = 1, \rho_{ij}: 1:2, 6:1, 11:2, 12:2, 19:2, 20:2, 24:1, 25:1, 32:1, 39:1$                              |                                                                                         |
| $w_i$                                                                                                               | $\langle -44t^3 - 40t^2 + 44t + 76,$<br>$-26t^3 - 74t^2 - 32t + 4 \rangle$              |
| $i = 16, D_i = 1, \rho_{ij}: 5:2, 7:1, 17:1, 24:1, 30:1$                                                            |                                                                                         |
| $w_i$                                                                                                               | $\langle t^3 + t^2 - 2,$<br>$t^3 + t - 2 \rangle$                                       |
| $i = 17, D_i = 1, \rho_{ij}: 3:1, 4:2, 5:1, 6:1, 11:1, 16:2, 26:2, 27:1$                                            |                                                                                         |
| $w_i$                                                                                                               | $\langle -2t^5 - 2t^4 + 4t^3 + 10t^2 + 10t + 4,$<br>$6t^4 + 12t^3 + 12t^2 + 6t \rangle$ |
| $i = 18, D_i = 1, \rho_{ij}: 0:2, 1:2, 3:1, 5:2, 6:2, 9:2, 13:1, 14:1, 23:1, 24:1, 28:1, 36:1, 37:2, 40:2, 43:2$    |                                                                                         |
| $w_i$                                                                                                               | $\langle 26t^3 - 56t - 78,$<br>$-8t^3 - 36t^2 - 64t - 96 \rangle$                       |
| $i = 19, D_i = 1, \rho_{ij}: 0:2, 1:1, 3:1, 12:1, 15:1, 28:2, 29:2, 32:2, 42:1, 43:2$                               |                                                                                         |
| $w_i$                                                                                                               | $\langle -2t^2 + 16t + 18,$<br>$6t^3 + 8t^2 + 14t + 12 \rangle$                         |
| $i = 20, D_i = 1, \rho_{ij}: 13:1, 14:2, 15:1, 22:2, 32:2, 37:1, 38:2, 41:1$                                        |                                                                                         |
| $w_i$                                                                                                               | $\langle -2t^3 + 10t^2 - 4t + 16,$<br>$-6t^3 + 8t^2 + 2t \rangle$                       |
| $i = 21, D_i = 1, \rho_{ij}: 3:2, 10:1, 12:2, 35:1, 38:1$                                                           |                                                                                         |
| $w_i$                                                                                                               | $\langle 0,$<br>$t^3 + 2t^2 - 3 \rangle$                                                |
| $i = 22, D_i = 1, \rho_{ij}: 20:1, 25:2, 28:1, 32:2, 37:1, 39:2, 40:1$                                              |                                                                                         |
| $w_i$                                                                                                               | $\langle -4t^2 + 14t + 6,$<br>$-2t^2 + 26 \rangle$                                      |
| $i = 23, D_i = 1, \rho_{ij}: 1:1, 2:2, 3:2, 13:1, 18:2, 24:1, 33:2, 40:1$                                           |                                                                                         |
| $w_i$                                                                                                               | $\langle -6t^3 - 6t^2 - 4t + 8,$<br>$-4t^3 - 2t^2 - 4t \rangle$                         |
| $i = 24, D_i = 1, \rho_{ij}: 9:1, 13:1, 15:2, 16:2, 18:2, 23:2, 27:2, 29:1, 30:1, 31:2, 36:1, 39:2, 40:1$           |                                                                                         |
| $w_i$                                                                                                               | $\langle 232t^2 - 528t + 920,$<br>$96t^2 - 232t + 1016 \rangle$                         |
| $i = 25, D_i = 1, \rho_{ij}: 0:1, 2:2, 4:2, 6:1, 14:2, 15:2, 22:1, 26:2, 32:2, 34:1, 39:1, 41:1$                    |                                                                                         |

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|                                                                                                  |                                                                                |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| $w_i$                                                                                            | $\langle -4t^3 - 2t^2 - 6t - 6, \\ -2t^3 - 8t - 2 \rangle$                     |
| $i = 26, D_i = 1, \rho_{ij}: 2:2, 12:1, 14:2, 17:1, 25:1, 34:1$                                  |                                                                                |
| $w_i$                                                                                            | $\langle 2t^3 + 4t^2 + 2t, \\ t^3 + 4t^2 + 4t + 1 \rangle$                     |
| $i = 27, D_i = 1, \rho_{ij}: 0:1, 2:1, 3:2, 10:1, 17:2, 24:1, 30:1, 31:1, 32:1, 34:1$            |                                                                                |
| $w_i$                                                                                            | $\langle 4t^3 + 16t^2 + 2t, \\ 10t^3 + 12t^2 - 10t - 16 \rangle$               |
| $i = 28, D_i = 1, \rho_{ij}: 4:1, 5:1, 18:2, 19:1, 22:2, 39:2$                                   |                                                                                |
| $w_i$                                                                                            | $\langle -6t - 4, \\ 2t^2 - 4 \rangle$                                         |
| $i = 29, D_i = 1, \rho_{ij}: 0:1, 2:2, 3:2, 5:2, 9:1, 19:1, 24:2, 31:2, 37:2, 41:1$              |                                                                                |
| $w_i$                                                                                            | $\langle 28t^3 + 60t^2 - 84t + 48, \\ 44t^3 - 24t^2 - 72t + 60 \rangle$        |
| $i = 30, D_i = 1, \rho_{ij}: 10:1, 16:2, 24:2, 27:2, 32:1, 33:2, 36:1$                           |                                                                                |
| $w_i$                                                                                            | $\langle 2t^3 + 8t^2, \\ -2t^3 + 4t^2 + 6t \rangle$                            |
| $i = 31, D_i = 1, \rho_{ij}: 3:1, 5:2, 12:2, 14:1, 24:1, 27:2, 29:1$                             |                                                                                |
| $w_i$                                                                                            | $\langle -2t^3 - 8t^2 + 4t + 6, \\ 2t^3 - 4t^2 - 4t + 6 \rangle$               |
| $i = 32, D_i = 1, \rho_{ij}: 7:1, 15:2, 19:1, 20:1, 22:1, 25:1, 27:2, 30:2, 33:1, 41:2$          |                                                                                |
| $w_i$                                                                                            | $\langle 14t^2 + 8t - 4, \\ -2t^4 - 4t^3 + 6t^2 + 8t + 4 \rangle$              |
| $i = 33, D_i = 1, \rho_{ij}: 0:2, 23:1, 30:1, 32:2, 35:1$                                        |                                                                                |
| $w_i$                                                                                            | $\langle 2t^2 - 2t + 2, \\ 2t^2 - 2t + 2 \rangle$                              |
| $i = 34, D_i = 1, \rho_{ij}: 25:2, 26:2, 27:2, 36:2, 39:1$                                       |                                                                                |
| $w_i$                                                                                            | $\langle 2t^2 - 8t + 8, \\ -2t + 4 \rangle$                                    |
| $i = 35, D_i = 1, \rho_{ij}: 3:2, 5:2, 10:1, 12:1, 21:2, 33:2, 38:2, 40:1$                       |                                                                                |
| $w_i$                                                                                            | $\langle 6t^3 + 12t^2 + 6t + 12, \\ 12t^2 + 18t + 24 \rangle$                  |
| $i = 36, D_i = 1, \rho_{ij}: 1:2, 3:2, 4:2, 5:2, 14:2, 18:2, 24:2, 30:2, 34:1$                   |                                                                                |
| $w_i$                                                                                            | $\langle -24t^3 - 38t^2 - 26t + 22, \\ -12t^3 - 22t^2 - 34t - 10 \rangle$      |
| $i = 37, D_i = 1, \rho_{ij}: 2:2, 3:2, 7:2, 8:1, 12:2, 18:1, 20:2, 22:2, 29:1$                   |                                                                                |
| $w_i$                                                                                            | $\langle -4t^3 - 8t^2 + 2t + 10, \\ -7t^3 - 20t^2 - 18t + 3 \rangle$           |
| $i = 38, D_i = 1, \rho_{ij}: 2:1, 5:1, 10:1, 12:2, 14:1, 20:1, 21:2, 35:1$                       |                                                                                |
| $w_i$                                                                                            | $\langle 2t^3 - 4t - 6, \\ 2t^3 - 4t^2 - 2t - 6 \rangle$                       |
| $i = 39, D_i = 1, \rho_{ij}: 4:1, 15:2, 22:1, 24:1, 25:2, 28:1, 34:2$                            |                                                                                |
| $w_i$                                                                                            | $\langle 4, \\ 4 \rangle$                                                      |
| $i = 40, D_i = 1, \rho_{ij}: 1:1, 2:2, 3:2, 6:2, 11:1, 12:2, 13:2, 18:1, 22:2, 23:2, 24:2, 35:2$ |                                                                                |
| $w_i$                                                                                            | $\langle -22t^3 + 84t^2 + 142t + 132, \\ -98t^3 - 240t^2 - 280t - 180 \rangle$ |
| $i = 41, D_i = 1, \rho_{ij}: 7:2, 11:1, 20:2, 25:2, 29:2, 32:1$                                  |                                                                                |
| $w_i$                                                                                            | $\langle 4t^2 + 4t, \\ 4t^2 - 4 \rangle$                                       |
| $i = 42, D_i = 1, \rho_{ij}: 1:1, 7:1, 9:1, 13:2, 19:2$                                          |                                                                                |
| $w_i$                                                                                            | $\langle -t^2 - t, \\ -t^2 + 1 \rangle$                                        |
| $i = 43, D_i = 1, \rho_{ij}: 2:1, 4:1, 5:1, 10:1, 18:1, 19:1$                                    |                                                                                |
| $w_i$                                                                                            | $\langle 0, \\ -t^3 + 1 \rangle$                                               |

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Table 3: The edges  $(i, j)$  with a linear endpoint, the linear endpoint  $k$  used, and the norm of  $R_{ij} = \text{Res}_t(f_k, g_{kl})$ .

| $(i, j)$  | $k$ | $N(R_{ij})$             | $(i, j)$   | $k$ | $N(R_{ij})$     |
|-----------|-----|-------------------------|------------|-----|-----------------|
| $(0, 7)$  | 0   | $3^3$                   | $(12, 14)$ | 12  | $2^4 \cdot 3$   |
| $(0, 13)$ | 0   | $2^4 \cdot 3^2 \cdot 7$ | $(12, 15)$ | 12  | $2^2 \cdot 7$   |
| $(0, 18)$ | 0   | $2^4 \cdot 3^4$         | $(12, 19)$ | 12  | $2^2 \cdot 3^2$ |
| $(0, 19)$ | 0   | $2^4 \cdot 3^2$         | $(12, 21)$ | 12  | $2^4$           |

| $(i, j)$ | $k$ | $N(R_{ij})$              | $(i, j)$ | $k$ | $N(R_{ij})$                    |
|----------|-----|--------------------------|----------|-----|--------------------------------|
| (0, 25)  | 0   | $2^4$                    | (12, 26) | 12  | $2^2$                          |
| (0, 27)  | 0   | $2^4 \cdot 3^2$          | (12, 31) | 12  | $2^2 \cdot 3^2 \cdot 7$        |
| (0, 29)  | 0   | $3 \cdot 7$              | (12, 35) | 12  | $3^3 \cdot 7$                  |
| (0, 33)  | 0   | $2^2 \cdot 3$            | (12, 37) | 12  | $2^2 \cdot 7$                  |
| (1, 8)   | 1   | $2^2 \cdot 3^5$          | (12, 38) | 12  | $2^2 \cdot 3^2 \cdot 13$       |
| (1, 12)  | 1   | $2^2$                    | (12, 40) | 12  | $7^2$                          |
| (1, 15)  | 1   | $3 \cdot 13$             | (13, 18) | 13  | $2^2 \cdot 3 \cdot 13$         |
| (1, 18)  | 1   | $2^4 \cdot 3$            | (13, 20) | 13  | $2^2 \cdot 3^3$                |
| (1, 19)  | 1   | $2^2 \cdot 3$            | (13, 23) | 13  | $2^8 \cdot 3$                  |
| (1, 23)  | 1   | $2^4 \cdot 3^4$          | (13, 24) | 13  | $2^2 \cdot 7^2$                |
| (1, 36)  | 1   | $2^2 \cdot 13^2$         | (13, 40) | 13  | $2^2 \cdot 3 \cdot 13$         |
| (1, 40)  | 1   | $2^8 \cdot 3^2$          | (13, 42) | 13  | $2^2 \cdot 3 \cdot 7$          |
| (1, 42)  | 1   | $2^2$                    | (14, 18) | 14  | $2^4 \cdot 3^2$                |
| (2, 8)   | 2   | $2^2$                    | (14, 20) | 14  | $2^2 \cdot 7^2$                |
| (2, 9)   | 2   | $2^2 \cdot 7$            | (14, 25) | 14  | $2^2 \cdot 7$                  |
| (2, 23)  | 2   | $2^2 \cdot 3$            | (14, 26) | 14  | $2^8 \cdot 7$                  |
| (2, 25)  | 2   | $2^2$                    | (14, 31) | 14  | $2^4 \cdot 3^4$                |
| (2, 26)  | 2   | $2^4 \cdot 3$            | (14, 36) | 14  | $2^2 \cdot 3$                  |
| (2, 27)  | 2   | $2^2 \cdot 7$            | (14, 38) | 14  | $3^3 \cdot 7$                  |
| (2, 29)  | 2   | $2^6$                    | (15, 19) | 15  | $2^2 \cdot 3$                  |
| (2, 37)  | 2   | $2^2 \cdot 3$            | (15, 20) | 15  | $2^8$                          |
| (2, 38)  | 2   | $3$                      | (15, 24) | 15  | $2^2 \cdot 3 \cdot 7 \cdot 13$ |
| (2, 40)  | 2   | $2^4 \cdot 3$            | (15, 25) | 15  | $2^2 \cdot 7$                  |
| (2, 43)  | 2   | $2^4$                    | (15, 32) | 15  | $2^4 \cdot 3^2$                |
| (3, 8)   | 3   | $2^6 \cdot 7$            | (15, 39) | 15  | $2^4$                          |
| (3, 10)  | 3   | $2^2 \cdot 13$           | (16, 17) | 16  | $2^6 \cdot 3^3 \cdot 7$        |
| (3, 17)  | 3   | $2^6$                    | (16, 24) | 16  | $2^4 \cdot 3 \cdot 7$          |
| (3, 18)  | 3   | $2^2 \cdot 3^2$          | (16, 30) | 16  | $3^4 \cdot 7^2$                |
| (3, 19)  | 3   | $2^2 \cdot 3 \cdot 7$    | (17, 26) | 17  | $2^4 \cdot 3^2$                |
| (3, 21)  | 3   | $2^4$                    | (17, 27) | 17  | $2^6 \cdot 3$                  |
| (3, 23)  | 3   | $2^6 \cdot 7$            | (18, 23) | 18  | $2^2 \cdot 3$                  |
| (3, 27)  | 3   | $2^2 \cdot 7$            | (18, 24) | 18  | $2^4 \cdot 3 \cdot 7$          |
| (3, 29)  | 3   | $2^4 \cdot 3$            | (18, 28) | 18  | $2^2$                          |
| (3, 31)  | 3   | $2^4 \cdot 3 \cdot 7$    | (18, 36) | 18  | $2^2 \cdot 3 \cdot 7$          |
| (3, 35)  | 3   | $2^2 \cdot 3^2 \cdot 13$ | (18, 37) | 18  | $2^2 \cdot 3$                  |
| (3, 36)  | 3   | $2^4 \cdot 3$            | (18, 40) | 18  | $2^6 \cdot 3 \cdot 13$         |
| (3, 37)  | 3   | $2^2$                    | (18, 43) | 18  | $2^4$                          |
| (3, 40)  | 3   | $2^4 \cdot 7$            | (19, 28) | 19  | $2^2 \cdot 7$                  |
| (4, 8)   | 4   | $2^2$                    | (19, 29) | 19  | $2^4 \cdot 3^2$                |
| (4, 14)  | 4   | $2^6$                    | (19, 32) | 19  | $2^4 \cdot 3^4$                |
| (4, 17)  | 4   | $2^2 \cdot 3$            | (19, 42) | 19  | $2^4 \cdot 3$                  |
| (4, 25)  | 4   | $2^2 \cdot 7$            | (19, 43) | 19  | $2^4$                          |
| (4, 28)  | 4   | $2^2 \cdot 7$            | (20, 22) | 20  | $2^2 \cdot 7^2$                |
| (4, 36)  | 4   | $2^2 \cdot 3^2$          | (20, 32) | 20  | $2^4 \cdot 3^3 \cdot 7$        |
| (4, 39)  | 4   | $2^2$                    | (20, 37) | 20  | $2^4 \cdot 3$                  |
| (4, 43)  | 4   | $2^4$                    | (20, 38) | 20  | $2^2 \cdot 3 \cdot 7$          |
| (5, 8)   | 5   | $7^2$                    | (20, 41) | 20  | $2^6$                          |
| (5, 16)  | 5   | $2^2 \cdot 3^3 \cdot 7$  | (21, 35) | 21  | $2^2 \cdot 3^2$                |
| (5, 17)  | 5   | $2^4 \cdot 3 \cdot 7$    | (21, 38) | 21  | $2^2 \cdot 3^2$                |
| (5, 18)  | 5   | $2^2 \cdot 7$            | (22, 25) | 22  | $2^2 \cdot 7^2$                |
| (5, 28)  | 5   | $2^2 \cdot 7$            | (22, 28) | 22  | $7^3$                          |
| (5, 29)  | 5   | $2^2 \cdot 7$            | (22, 32) | 22  | $2^4 \cdot 3^3 \cdot 7$        |
| (5, 31)  | 5   | $7^2$                    | (22, 37) | 22  | $3^5$                          |
| (5, 35)  | 5   | $2^2 \cdot 3 \cdot 7$    | (22, 39) | 22  | $2^6$                          |
| (5, 36)  | 5   | $3 \cdot 7$              | (22, 40) | 22  | $2^4 \cdot 3 \cdot 7$          |
| (5, 38)  | 5   | $2^2 \cdot 3 \cdot 7$    | (23, 24) | 23  | $2^8 \cdot 3$                  |
| (5, 43)  | 5   | $2^6$                    | (23, 33) | 23  | $2^4$                          |

| $(i, j)$ | $k$ | $N(R_{ij})$             | $(i, j)$ | $k$ | $N(R_{ij})$             |
|----------|-----|-------------------------|----------|-----|-------------------------|
| (6, 9)   | 6   | $2^4$                   | (23, 40) | 23  | $2^4 \cdot 3^5 \cdot 7$ |
| (6, 15)  | 6   | $3^4$                   | (24, 27) | 24  | $2^2 \cdot 3^2 \cdot 7$ |
| (6, 17)  | 6   | $2^2 \cdot 3$           | (24, 29) | 24  | $2^2 \cdot 7^2$         |
| (6, 18)  | 6   | $2^2 \cdot 3$           | (24, 30) | 24  | $2^2 \cdot 3^2 \cdot 7$ |
| (6, 25)  | 6   | $2^4$                   | (24, 31) | 24  | $2^2 \cdot 7^3$         |
| (6, 40)  | 6   | $2^6 \cdot 3$           | (24, 36) | 24  | $2^2 \cdot 7 \cdot 13$  |
| (7, 16)  | 7   | $2^2 \cdot 3^4$         | (24, 39) | 24  | $2^6$                   |
| (7, 32)  | 7   | $3^3$                   | (24, 40) | 24  | $2^2 \cdot 3 \cdot 7$   |
| (7, 37)  | 7   | $2^4 \cdot 3$           | (25, 26) | 25  | $2^2 \cdot 7$           |
| (7, 41)  | 7   | $2^2 \cdot 3^2$         | (25, 32) | 25  | $2^4 \cdot 7^2$         |
| (7, 42)  | 7   | $2^2 \cdot 3^2$         | (25, 34) | 25  | $2^4$                   |
| (8, 10)  | 8   | $2^4$                   | (25, 39) | 25  | $2^4$                   |
| (8, 12)  | 8   | $2^6 \cdot 7$           | (25, 41) | 25  | $2^2 \cdot 7$           |
| (8, 37)  | 8   | $2^2 \cdot 3$           | (26, 34) | 26  | $2^2 \cdot 3^2$         |
| (9, 12)  | 9   | $2^4 \cdot 7$           | (27, 30) | 27  | $3^4$                   |
| (9, 13)  | 9   | $2^2 \cdot 7^3$         | (27, 31) | 27  | $2^2 \cdot 7$           |
| (9, 18)  | 9   | $2^2 \cdot 7$           | (27, 32) | 27  | $2^8 \cdot 3 \cdot 7$   |
| (9, 24)  | 9   | $7^3$                   | (27, 34) | 27  | $3^4$                   |
| (9, 29)  | 9   | $2^4$                   | (28, 39) | 28  | $2^2$                   |
| (9, 42)  | 9   | $2^2 \cdot 7$           | (29, 31) | 29  | $2^2 \cdot 3^5$         |
| (10, 12) | 10  | $2^2 \cdot 13$          | (29, 37) | 29  | $2^8 \cdot 7$           |
| (10, 13) | 10  | $2^2 \cdot 3$           | (29, 41) | 29  | $2^6 \cdot 3^2$         |
| (10, 21) | 10  | $2^2 \cdot 3^2$         | (30, 32) | 30  | $2^4 \cdot 3^2$         |
| (10, 27) | 10  | $2^4$                   | (30, 33) | 30  | $2^4$                   |
| (10, 30) | 10  | $2^2 \cdot 3$           | (30, 36) | 30  | $2^2 \cdot 3$           |
| (10, 35) | 10  | $2^2 \cdot 3 \cdot 13$  | (32, 33) | 32  | $2^2 \cdot 3$           |
| (10, 38) | 10  | $2^2 \cdot 3 \cdot 13$  | (32, 41) | 32  | $2^2 \cdot 3^3$         |
| (10, 43) | 10  | $2^4$                   | (33, 35) | 33  | $2^2 \cdot 3^2$         |
| (11, 15) | 11  | $2^2 \cdot 3^2 \cdot 7$ | (34, 36) | 34  | $2^4$                   |
| (11, 17) | 11  | $2^6 \cdot 3^3$         | (34, 39) | 34  | $2^4$                   |
| (11, 40) | 11  | $2^2 \cdot 3^2$         | (35, 38) | 35  | $2^2 \cdot 3^6$         |
| (11, 41) | 11  | $2^2 \cdot 3 \cdot 7$   | (35, 40) | 35  | 3                       |
| (12, 13) | 12  | $2^2 \cdot 3 \cdot 7^2$ |          |     |                         |

Table 4: The terminal residue classes  $x \equiv a, y \equiv b \pmod{p^k}$  of Step 4, grouped by  $p, k$  and the value of  $B_p$ .

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                        |
|-----|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | 2   | 2     | (1, 3), (3, 1)                                                                                                                                                                                                                                                          |
| 2   | 3   | 2     | (0, 5), (2, 7), (4, 1), (6, 3)                                                                                                                                                                                                                                          |
| 3   | 3   | 2     | (0, 20), (2, 2), (3, 17), (5, 11), (6, 14), (9, 11), (10, 4), (11, 2), (12, 8), (14, 11), (15, 5), (18, 2), (19, 13), (20, 2), (21, 26), (23, 11), (24, 23)                                                                                                             |
| 3   | 4   | 2     | (1, 22), (4, 61), (7, 19), (13, 16), (16, 55), (17, 47), (22, 52), (25, 10), (26, 74), (28, 49), (31, 7), (34, 46), (40, 43), (43, 1), (44, 47), (49, 79), (52, 37), (53, 74), (55, 76), (58, 34), (61, 73), (67, 70), (70, 28), (71, 47), (76, 25), (79, 64), (80, 74) |
| 3   | 5   | 2     | (8, 101), (35, 182), (62, 20), (89, 101), (116, 182), (143, 20), (170, 101), (197, 182), (224, 20)                                                                                                                                                                      |
| 7   | 2   | 0     | (1, 25), (1, 28), (1, 45), (8, 4), (8, 21), (8, 24), (11, 41), (15, 3), (15, 14), (18, 6), (22, 7), (22, 11), (22, 31), (25, 20), (29, 10), (32, 34), (39, 48), (43, 17), (43, 35), (43, 46), (46, 13)                                                                  |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7   | 3   | 0     | (4, 27), (15, 32), (29, 137), (29, 196), (36, 116), (36, 234), (36, 336),<br>(53, 125), (64, 228), (78, 147), (78, 333), (85, 87), (85, 287), (85, 312),<br>(113, 81), (127, 98), (127, 186), (134, 165), (134, 238), (134, 283),<br>(151, 321), (162, 277), (176, 39), (176, 49), (183, 18), (183, 136),<br>(183, 189), (200, 76), (211, 130), (225, 0), (225, 235), (232, 140),<br>(232, 214), (232, 332), (249, 174), (260, 326), (274, 88), (274, 294),<br>(281, 67), (281, 91), (281, 185), (298, 272), (309, 179), (323, 245),<br>(323, 284), (330, 38), (330, 42), (330, 263) |
| 7   | 4   | 0     | (102, 2281), (445, 566), (788, 1252), (1131, 1938), (1474, 223),<br>(1817, 909), (2160, 1595)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 13  | 1   | 0     | (0, 3), (3, 8), (4, 3), (10, 3), (12, 3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 13  | 2   | 0     | (5, 99), (7, 23), (8, 118), (18, 151), (20, 127), (21, 14), (31, 34),<br>(33, 62), (34, 79), (44, 86), (46, 166), (47, 144), (57, 138), (59, 101),<br>(60, 40), (70, 21), (72, 36), (73, 105), (85, 140), (86, 1), (96, 125),<br>(98, 75), (99, 66), (109, 8), (111, 10), (112, 131), (122, 60), (124, 114),<br>(125, 27), (135, 112), (137, 49), (138, 92), (148, 164), (150, 153),<br>(151, 157), (161, 47), (163, 88), (164, 53)                                                                                                                                                  |
| 13  | 3   | 0     | (83, 1594), (252, 73), (421, 749), (590, 1425), (759, 2101), (928, 580),<br>(1097, 1256), (1266, 1932), (1435, 411), (1604, 1087), (1773, 1763),<br>(1942, 242), (2111, 918)                                                                                                                                                                                                                                                                                                                                                                                                         |